



Project Report

Statistics and Probability I

Statistical and Probabilistic Analysis of HDFS Logs for Anomaly Detection

Submitted by:

Aya ABID Farah AMHALI
Aya SQUALLI HOUSSAINI

Supervised by:

Prof. Ikram Chairi

Academic Year: 2025-2026

1 Choice of Topic and Data Source

In this project, we chose to study **anomaly detection in HDFS (Hadoop Distributed File System) logs**. This topic is relevant because system logs play a crucial role in monitoring, maintaining, and securing distributed systems.

The data used comes from the **Kaggle** platform, specifically the LogHub HDFS dataset (HDFS_v1). This dataset is widely used in the data science and machine learning community for log anomaly detection research.

- **Dataset name:** HDFS Log Dataset (LogHub HDFS_v1)
- **Source:** Kaggle (ayenuryrr/loghub-hdfs-hadoop-distributed-file-system-data)
- **Dataset objective:** Identify normal and abnormal behaviors in HDFS logs
- **Anomaly rate:** 2.93% (16,838 anomalies out of 575,061 total instances)

The dataset contains four main files:

- `anomaly_label.csv`: BlockId with labels (Normal/Anomaly)
- `Event_occurrence_matrix.csv`: Event occurrence counts by BlockId
- `Event_traces.csv`: Detailed event sequences
- `HDFS.log_templates.csv`: Event template definitions

The choice of this dataset is motivated by its clear structure, the presence of a **target variable (normal / anomaly)**, its relevance to real-world distributed systems, and its direct connection to concepts studied in statistics and probability courses.

2 Data Processing and Structuring

Before any statistical analysis, we properly prepared the data to avoid interpretation errors and ensure reliable results.

2.1 Data Loading and Merging

The dataset was loaded using the KaggleHub API. We merged the anomaly labels with the event occurrence matrix using `BlockId` as the key. The label was encoded as binary: 0 for normal and 1 for anomaly.

2.2 Data Cleaning and Feature Engineering

- Standardized column names and removed duplicates on `BlockId`
- Ensured consistent data types for `BlockId`
- Encoded labels: `Normal` $\rightarrow$ 0, `Anomaly` $\rightarrow$ 1
- Converted event counts to numeric, filling NaN with 0

- Engineered core features:
 - `total_count`: Sum of all event counts per block
 - `unique_events`: Number of distinct event types per block
 - `max_event_count`: Maximum count of any single event per block
 - `log_total`: Natural logarithm of `total_count` (log1p transformation)
- Selected top-15 most frequent events for analysis

2.3 Final Dataset Structure

The final feature table contains 575,061 instances with 21 features:

- `BlockId`: Unique identifier
- `y`: Target variable (0=normal, 1=anomaly)
- Core engineered features (4)
- Top-15 event columns by global frequency

Feature Engineering Insight

The log1p transformation ($\log(1 + x)$) was applied to `total_count` to handle its skewed distribution and reduce the influence of extreme values. This transformation is particularly useful for count data that may contain zeros.

3 Univariate Analysis

Univariate analysis focuses on studying each individual feature's distribution and relationship with the target variable.

3.1 Summary Statistics

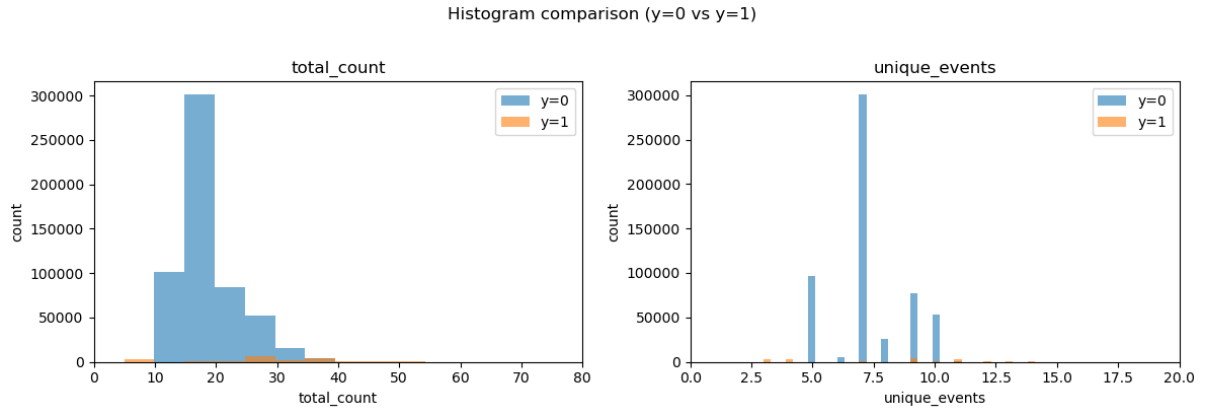
We computed comprehensive summary statistics including mean, standard deviation, skewness, kurtosis, and interquartile range (IQR) for all features. The table below shows key statistics for core features:

Table 1: Univariate Summary Statistics for Core Features

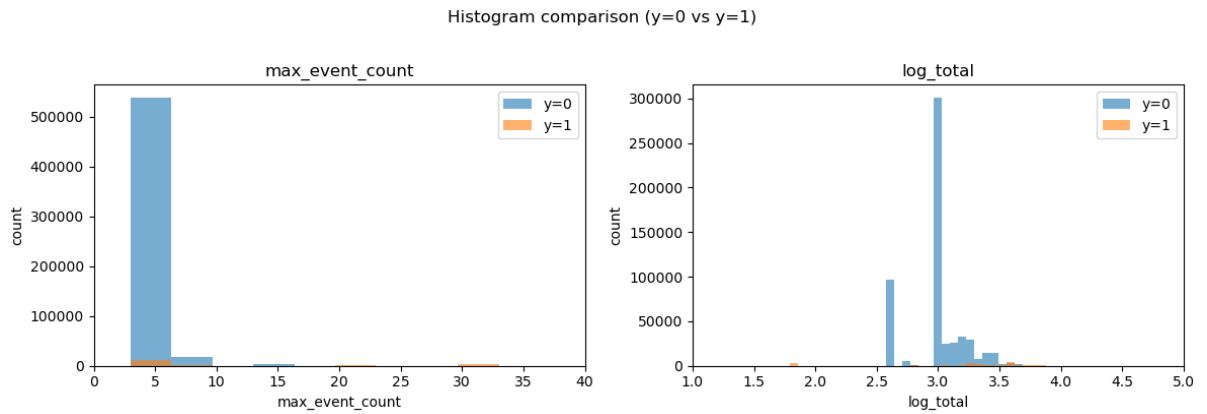
Feature	n	Mean	Std	Skew	Kurtosis	Outlier Rate (IQR)
<code>total_count</code>	575,061	19.708	5.286	38.855	2088.017	0.000
<code>unique_events</code>	575,061	7.277	1.569	1.415	4.387	0.003
<code>max_event_count</code>	575,061	3.591	3.117	41.347	2024.549	0.000
<code>log_total</code>	575,061	3.003	0.238	2.664	13.827	0.001

3.2 Distribution Analysis

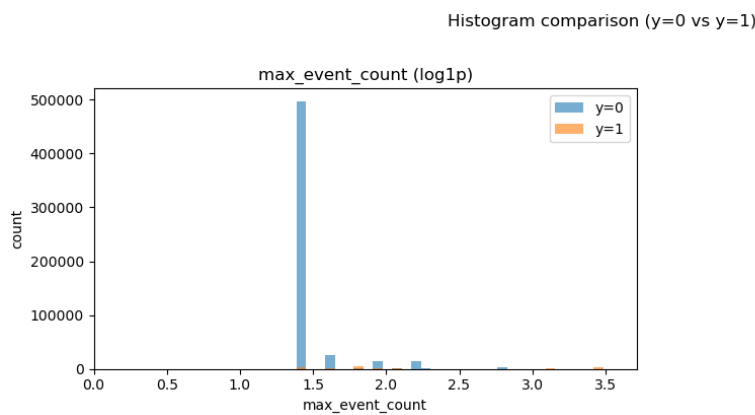
Histograms were plotted for all core features comparing normal ($y=0$) and anomalous ($y=1$) instances. The distributions reveal several important patterns:



(a) Histogram: total_count and unique_events(normal vs anomaly)



(b) Histogram: max_event_count and log_total (normal vs anomaly)



(c) Histogram: log(max_total) (normal vs anomaly)

Figure 1: Histogram comparisons for core features

3.3 Empirical Cumulative Distribution Functions (ECDF)

ECDF plots provide another perspective on distribution differences between classes:

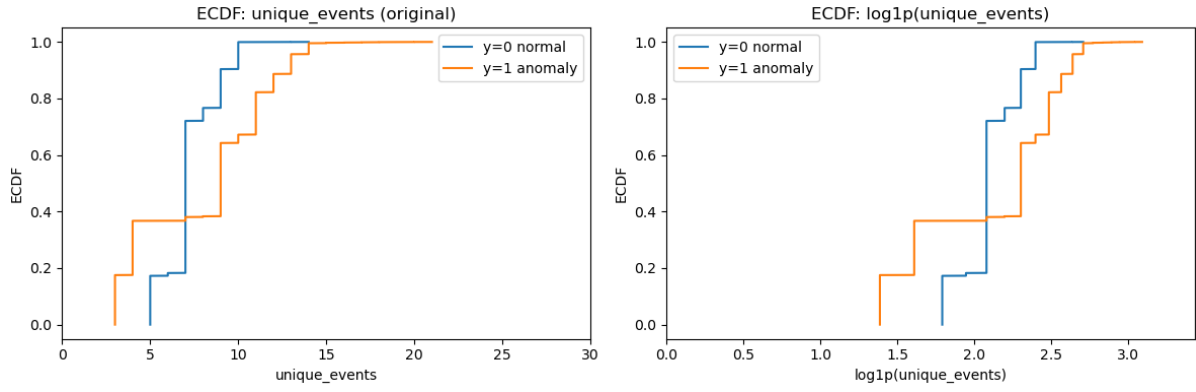
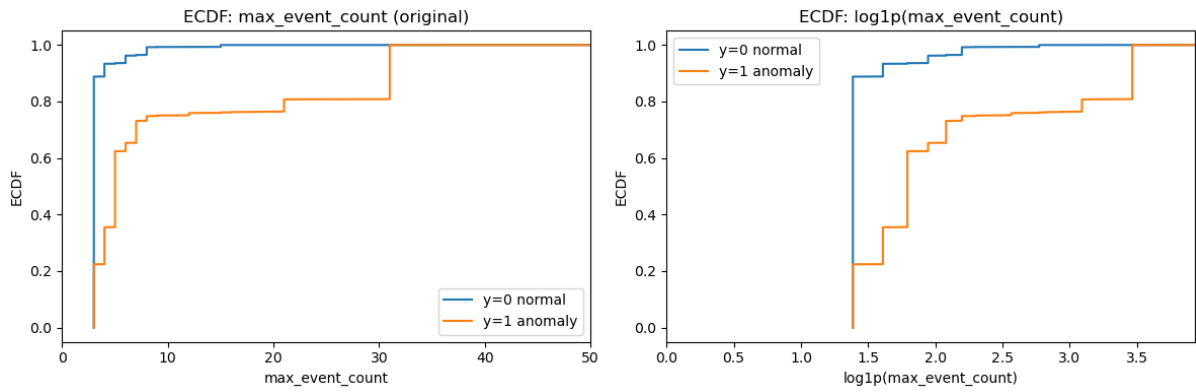
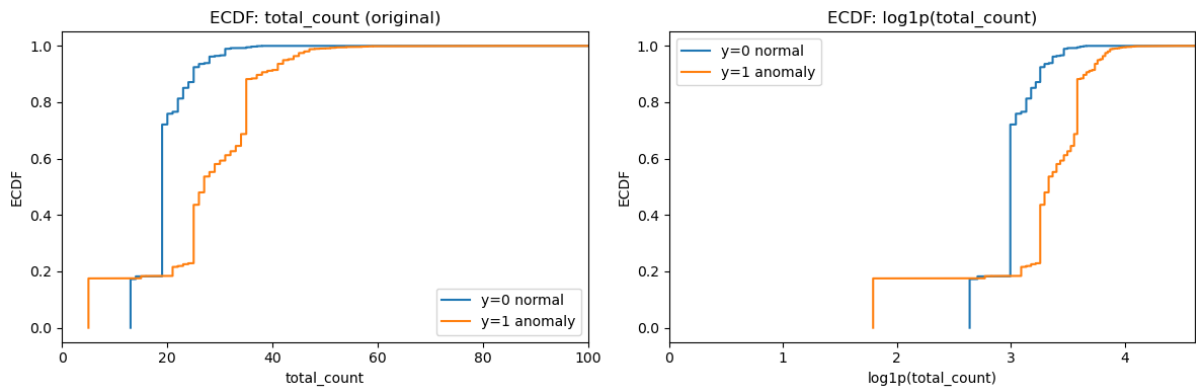
(a) ECDF comparison for `unique_events`(b) ECDF comparison for `max_event_count`(c) ECDF comparison for `total_count`

Figure 2: ECDF plots for core features (original and log-transformed)

4 Multivariate Analysis

Multivariate analysis studies relationships between multiple features simultaneously, helping us understand how feature combinations distinguish normal from anomalous logs.

4.1 Correlation Analysis

The correlation matrix reveals relationships between features:

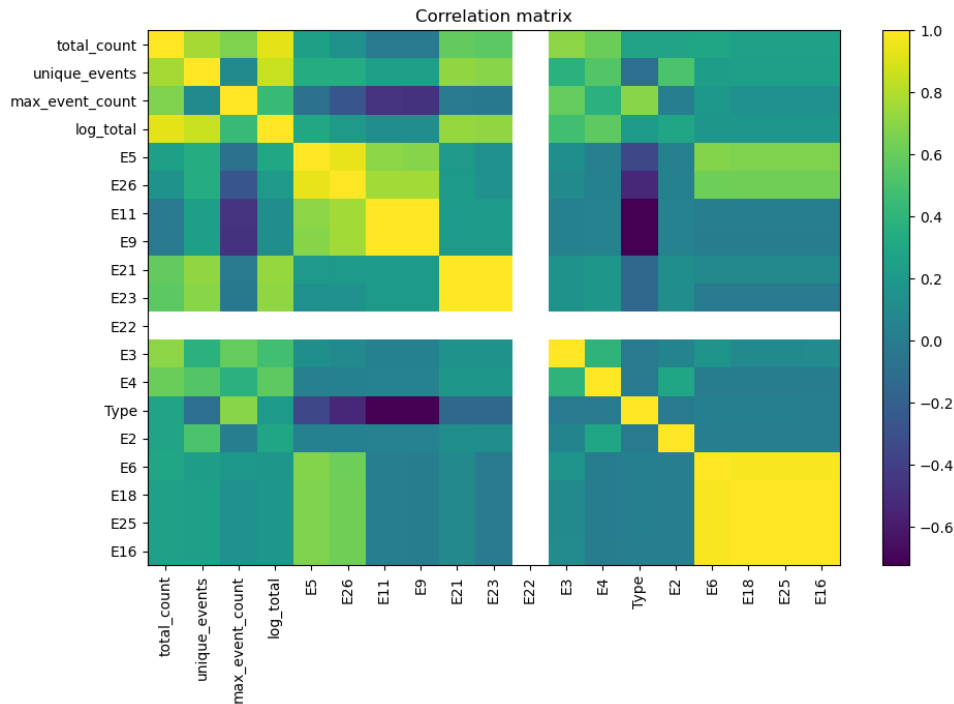


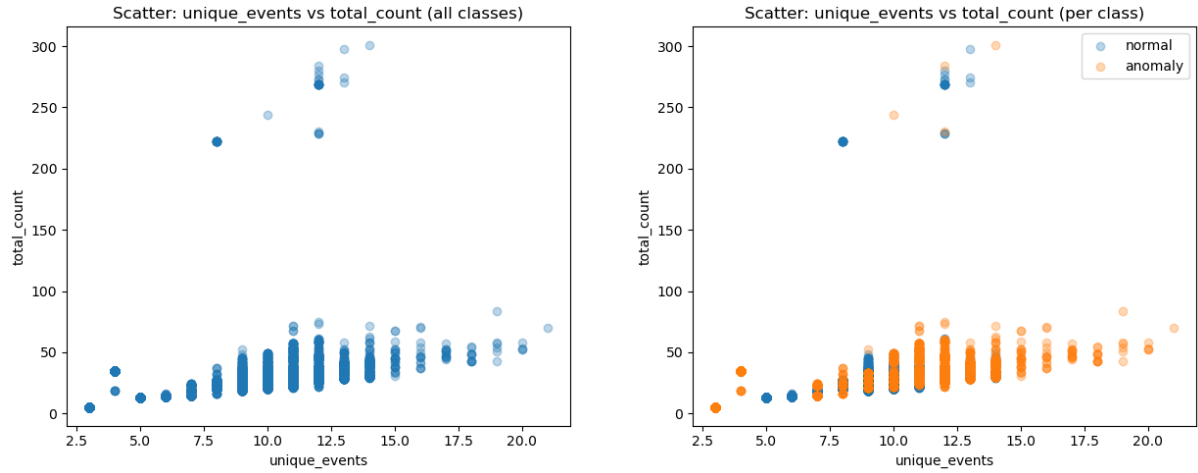
Figure 3: Correlation matrix of core features and top events

Key observations:

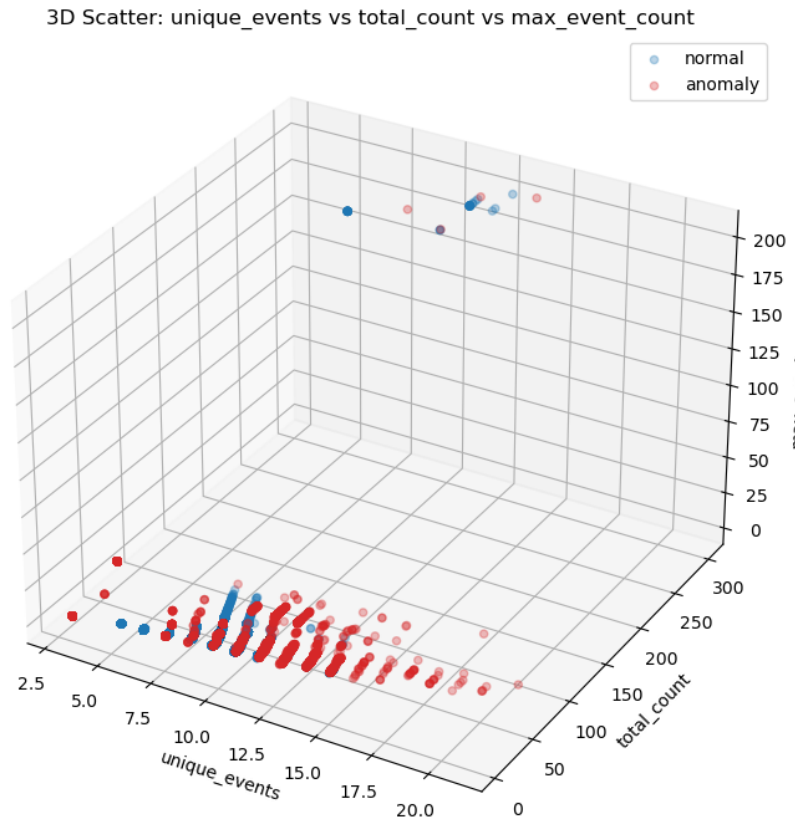
- total_count and unique_events show strong positive correlation (expected)
- log_total shows moderate correlation with other features
- Some event types show strong correlations with each other

4.2 Scatter Plot Analysis

2D and 3D scatter plots reveal relationships between features:



(a) unique_events vs total_count (all classes) (b) unique_events vs total_count (by class)



(c) 3D scatter: unique_events vs total_count vs max_event_count

Figure 4: Scatter plot analyses

4.3 Derived Metrics

We computed several derived metrics to capture block behavior:

- average_repetitions: $\text{total_count} / \text{unique_events}$
- density_score: $\text{unique_events} / \text{total_count}$

- `volume_diversity_product`: `total_count` $\times$ `unique_events`

5 Study of Distributions and Probability Laws

We investigated whether the empirical distributions of our features follow known probability laws, with particular focus on count data distributions.

5.1 Distribution Fitting for `max_event_count`

The `max_event_count` feature showed interesting distributional properties. After removing a significant outlier (value=203), we fitted Poisson and Negative Binomial distributions:

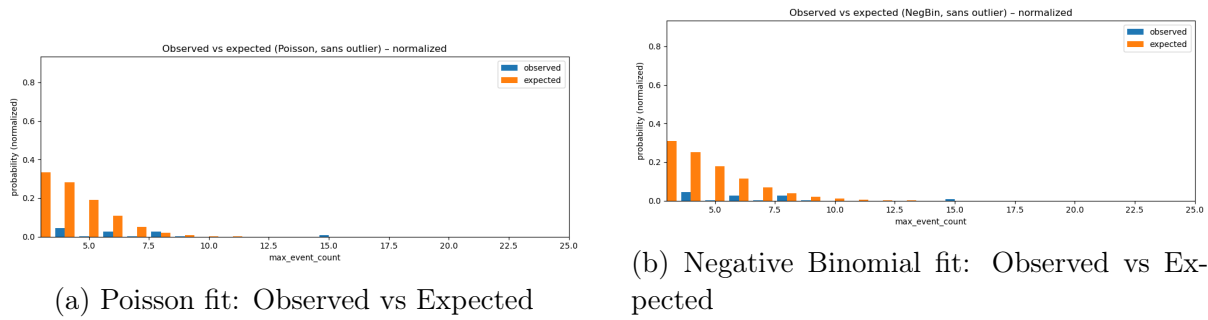


Figure 5: Distribution fitting for `max_event_count`

5.2 Goodness-of-Fit Assessment

We used p-values from fitted distributions to assess extremity of values:

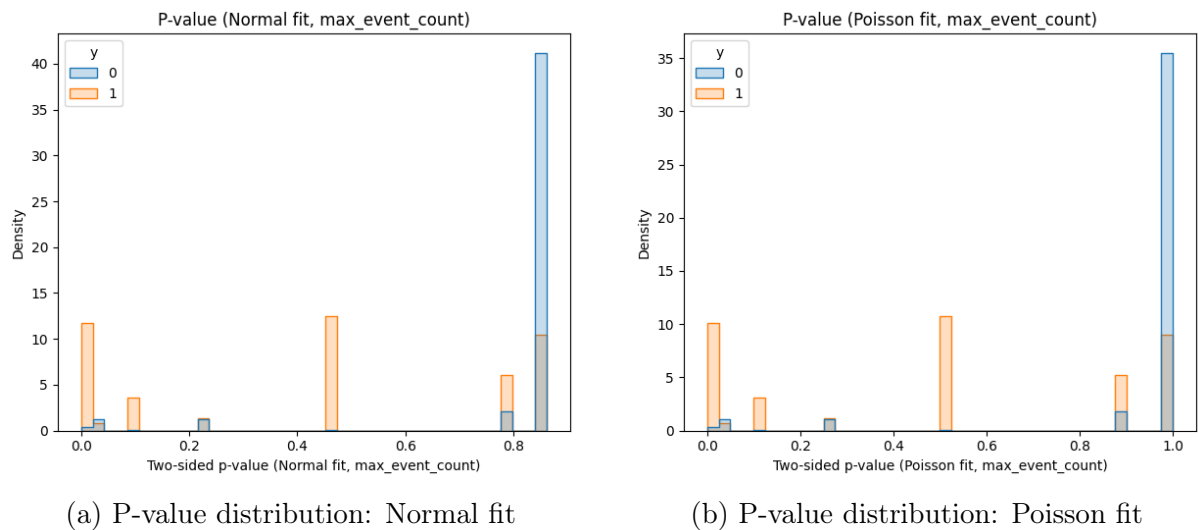


Figure 6: P-value distributions for anomaly detection

Key findings:

- 24.96% of anomalies have p-value < 0.01 for both Normal and Poisson fits

- Only 0.73% of normal instances have p-value < 0.01
- This suggests p-values from distribution fits could serve as anomaly scores

5.3 Akaike Information Criterion (AIC) for Model Selection

We used AIC to determine the best-fitting distribution for each feature:

Table 2: Best Fit Distributions by AIC

Feature	Kind	Best Distribution	n	AIC
log_total	continuous	Student's t	575,061	-493,969.4
max_event_count	count	Poisson	575,061	2,345,804
total_count	count	Negative Binomial	575,061	3,448,171
unique_events	count	Poisson	575,061	2,392,806

6 Linear Regression Analysis

We performed linear regression on the normal class only to avoid overfitting from anomalous points, exploring relationships between various features.

6.1 Regression Models

6.1.1 log_total vs. unique_events

Model for normal instances ($y=0$):

$$\log_total = 2.031 + 0.133 \times \text{unique_events}, \quad R^2 = 0.8427$$

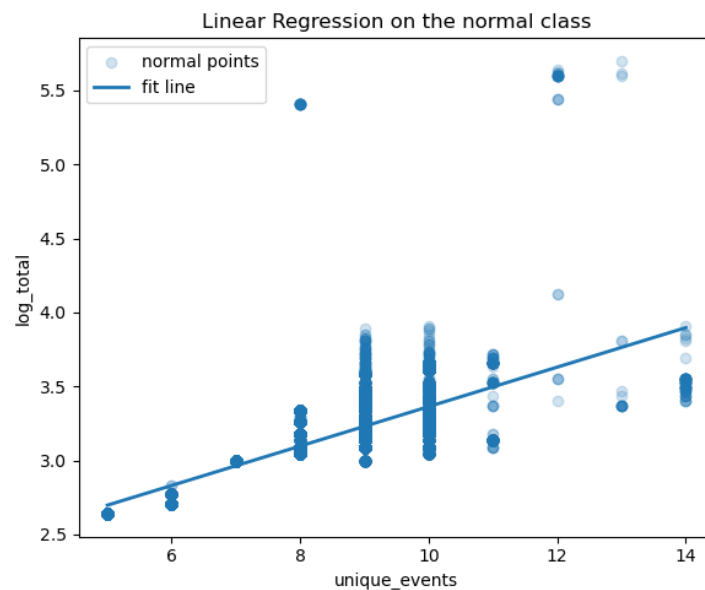


Figure 7: Linear Regression: log_total vs. unique_events (normal class)

6.1.2 total_count vs. unique_events

$$\text{total_count} = -0.190 + 2.714 \times \text{unique_events}, \quad R^2 = 0.6803$$

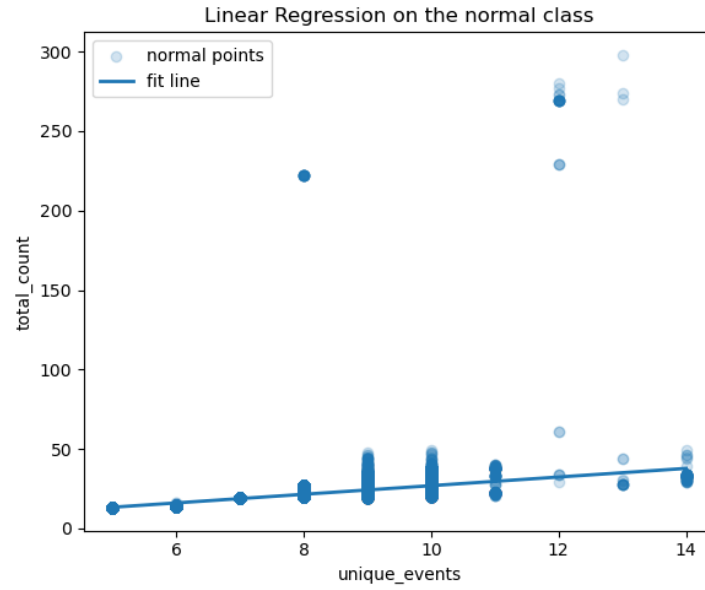


Figure 8: Linear Regression: total_count vs. unique_events (normal class)

6.1.3 max_event_count vs. unique_events

$$\text{max_event_count} = 0.738 + 0.365 \times \text{unique_events}, \quad R^2 = 0.0580$$

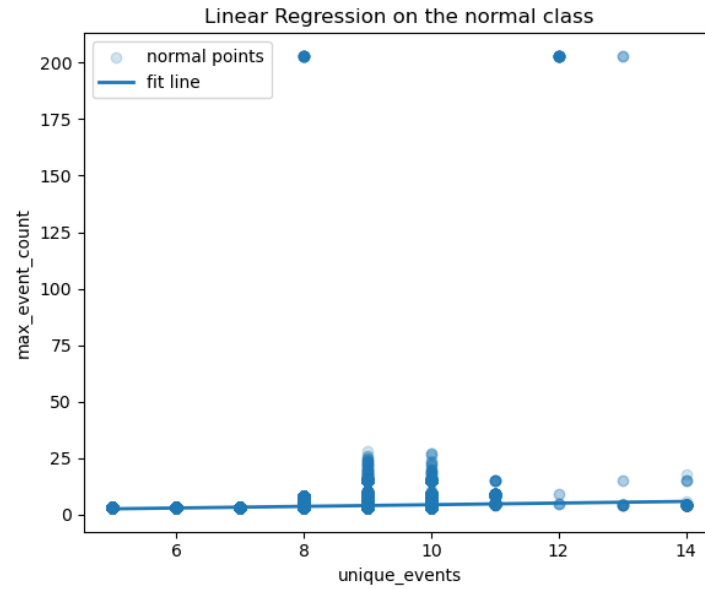


Figure 9: Linear Regression: max_event_count vs. unique_events (normal class)

6.1.4 total_count vs. max_event_count

$$\text{total_count} = 14.310 + 1.535 \times \text{max_event_count}, \quad R^2 = 0.4986$$

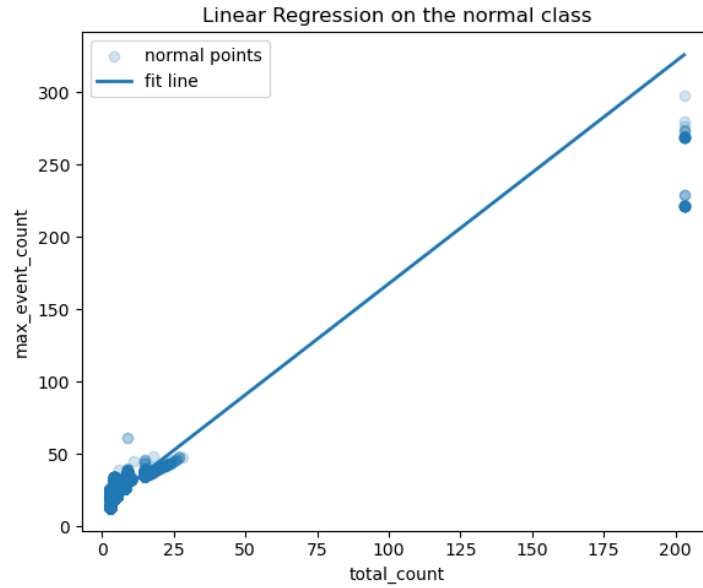


Figure 10: Linear Regression: total_count vs. max_event_count (normal class)

6.2 Observations

These linear regressions provide insight into typical trends within the normal class:

- **unique_events vs. log_total:** Strong linear relationship ($R^2 = 0.84$)
- **unique_events vs. total_count:** Moderate linear relationship ($R^2 = 0.68$)
- **unique_events vs. max_event_count:** Weak relationship ($R^2 = 0.06$)
- **max_event_count vs. total_count:** Moderate relationship ($R^2 = 0.50$)

6.3 Conclusion on Linear Regression

Although these models offer intuition about feature relationships in the normal class, the variance and nonlinearity observed limit the predictive accuracy of simple linear regression. More flexible or robust statistical methods may be required to better capture the relationships, especially for anomaly detection or rare events.

7 Logistic Regression for Anomaly Detection

We implemented a logistic regression model to classify normal vs. anomalous logs based on selected features.

7.1 Model Setup

- **Features:** Core engineered features + top-15 events
- **Preprocessing:** Standard scaling (mean=0, std=1)
- **Train/Test split:** 75%/25% with stratified sampling
- **Class weighting:** Balanced to handle class imbalance

7.2 Model Performance

The logistic regression achieved excellent performance:

Classification Report:

	precision	recall	f1-score	support
0	1.000	1.000	1.000	139556
1	0.995	1.000	0.997	4210
accuracy			1.000	143766
macro avg	0.998	1.000	0.999	143766
weighted avg	1.000	1.000	1.000	143766

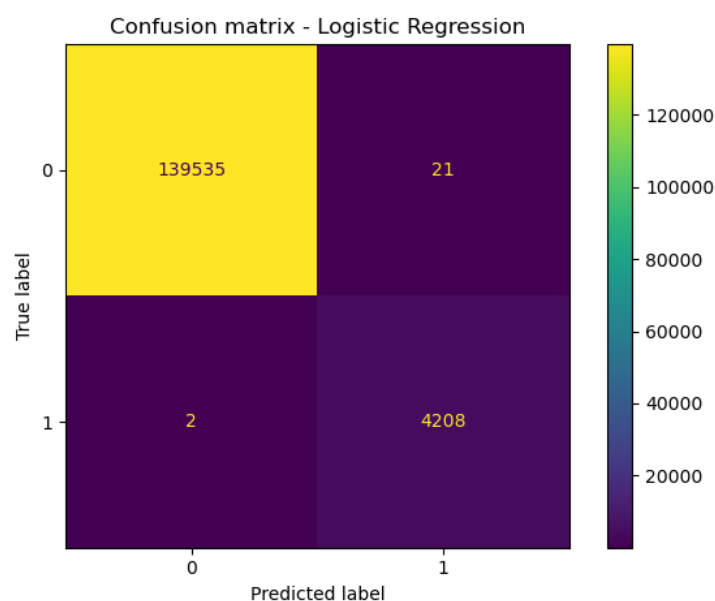


Figure 11: Confusion matrix for logistic regression

7.3 Feature Importance

The logistic regression coefficients reveal which features are most important for anomaly detection:

Table 3: Top Feature Importance from Logistic Regression

Feature	Coefficient (Absolute Value)
Type	25.66
total_count	17.30
E3	11.48
E4	6.78
unique_events	6.26
E23	4.50
max_event_count	4.15
E2	3.56

8 Interpretation of Results

Our comprehensive analysis reveals several key findings:

- **Distribution Differences:** Anomalous instances often exhibit extreme values and greater dispersion compared to normal instances
- **Feature Relationships:** Strong linear relationships exist between some features (e.g., `unique_events` and `total_count`) in the normal class
- **Distributional Properties:** The data shows overdispersion, making Negative Binomial distributions more appropriate than Poisson for count data
- **Anomaly Detection:** Simple statistical methods (distribution p-values) can identify anomalies, with 25% of anomalies having p-value < 0.01 vs. 0.7% of normals
- **Predictive Modeling:** Logistic regression achieves near-perfect classification with appropriate feature engineering and preprocessing

9 Conclusion

This project demonstrated a comprehensive statistical and probabilistic analysis of HDFS logs for anomaly detection. We followed a systematic approach:

1. **Data Preparation:** Imported, cleaned, and engineered features from raw HDFS logs
2. **Exploratory Analysis:** Conducted univariate and multivariate analysis to understand data distributions and relationships
3. **Distribution Fitting:** Investigated how well empirical distributions match theoretical probability laws
4. **Regression Analysis:** Explored linear relationships between features in the normal class
5. **Predictive Modeling:** Built and evaluated a logistic regression model for anomaly detection

Key insights from our analysis:

- Statistical methods provide a solid foundation for understanding log data patterns
- Feature engineering (especially derived metrics) significantly improves anomaly detection
- Distribution-based anomaly scoring offers a simple yet effective approach
- The class imbalance (2.93% anomalies) must be carefully handled in modeling

This work demonstrates that statistical tools can identify significant differences between normal and abnormal behaviors, providing a foundation for more advanced methods in machine learning and anomaly detection. Future work could explore more complex models, additional feature engineering, and real-time anomaly detection systems.